



MolRL: Self-supervised molecular image representation learning via graph structure bootstrapping

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ABSTRACT

Molecular Representation Learning offers great potential in efficiently predicting molecular properties and fostering the innovative design of functional and novel chemical compounds. Nevertheless, acquiring labeled molecule data is a costly and time-consuming endeavor, posing a significant challenge for supervised-learning models to generalize effectively across the giant chemical space. In this work, we present a self-supervised pretraining deep learning framework, named MolRL, pretrained on 10 million unlabelled molecular images, to learn molecular representation and predict molecular property. Our Molecular Image Representation Learning framework dissects molecular images into patches, leveraging graph structure bootstrapping with local Routing and global attention. A self-supervised contrastive learning approach co-trains anchor and learner views of augmented molecular graphs, exploiting CNNs' image feature extraction and hierarchical graph analysis for precise molecular feature representation. Experimental results demonstrate that our proposed Structure Bootstrapping Self-Supervision framework achieves remarkable accuracy across diverse molecular property benchmarks, encompassing both classification and regression tasks. Leveraging the power of pre-training on a vast unlabeled molecular database, MolRL exhibits an intriguing capability of clustering molecules based on their underlying labels, even without direct access to these labels during the training phase. This phenomenon underscores the framework's proficiency in efficiently extracting meaningful chemical structure information from the data, highlighting its potential for advancing molecular representation learning. The code for our proposed model is publicly available on the website: [GitHub/MolRL](https://github.com/MolRL).

1. Introduction

Molecular representation stands as a cornerstone in the innovative design of functional and novel chemical compounds, as it enables a comprehensive understanding of the intricate structures and properties of molecules. Given the staggering diversity and vastness of potential stable chemical compounds, crafting an informative and generalizable representation that encapsulates the entire chemical space poses a formidable challenge [1]. Historically, conventional molecular representations, such as Extended-Connectivity Fingerprints [2] (ECFP), have been instrumental in facilitating computational chemistry endeavors by providing a standardized framework for encoding molecular features. In recent years, the advent of machine learning methodologies has revolutionized the landscape of molecular representation learning. This data-driven approach harnesses the power of algorithms to automatically extract and learn meaningful representations from vast amounts of chemical data. These representations, tailored for specific tasks or capable

of broad generalization, have sparked significant interest in various applications, including but not limited to, accurate prediction of chemical properties, sophisticated chemical modeling, and innovative molecule design [3].

However, learning such representations poses formidable challenges, primarily stemming from three aspects. Firstly, the conventional string representations [4] adopted by classical machine learning and some deep learning methods, despite their merits, are inherently incapable of directly encoding the crucial topological structural information within molecules. Graph models [5], when confronted with large-scale and intricate drug molecular graphs, encounter issues related to computational complexity and efficiency, along with limited accuracy in feature description. ImageMol [3] ingeniously transforms molecular structures into image representations, leveraging computer vision and self-supervised learning techniques to extract profound semantic features from vast unlabeled molecular image datasets. Nonetheless, this approach, from the perspective of image texture features, overlooks the

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inherent graph-structured nature of molecules. Secondly, the vastness of chemical space, with an estimated number of potential pharmacologically active molecules reaching 10^{60} [6], underscores the paramount issue of data scarcity in molecular learning tasks. Experimental validation of molecular properties often necessitates intricate and time-consuming laboratory procedures, resulting in exorbitant costs for acquiring labeled data [6]. Consequently, machine learning models trained on such limited data are prone to overfitting, impairing their generalization capabilities to novel compound molecules. Thirdly, to preserve the abundant structural information, numerous recent endeavors have harnessed Graph Neural Networks (GNNs), demonstrating promising outcomes in molecular property prediction [7]. Nevertheless, the current self-supervised pre-training frameworks for GNNs predominantly concentrate on node-level or graph-level tasks. These methods fall short in capturing the rich information embedded within subgraphs or graph motifs. For instance, molecular functional groups (frequently occurring subgraphs in molecular graphs) typically carry indicative information about molecular properties, rendering the effective extraction of functional group information a crucial metric for assessing model performance.

In this study, we introduce a Molecular Image Representation Learning framework that innovatively dissects molecular images into patches, conceptualized as graph nodes, and subsequently engages in graph structure bootstrapping learning. This approach innovatively integrates a Graph Learner with a hybrid mechanism comprising local Routing and global graph attention, rooted in the principles of weighted graph isomorphism network and graph pooling functionality. This composite neural network architecture not only proposes a bootstrapping schema for the graph structure but also facilitates a hierarchical analysis for the chemical information, progressing from the atomic level, through chemical bonds, to the functional group level within the molecular structure. To train this model, we devise a self-supervised framework that leverages a contrastive learning strategy to co-train both the anchor view and learner view of the augmented molecular representations. The principal advantage of our model resides in its integration within a self-supervised framework, which harnesses the formidable capability of convolutional neural networks (CNNs) to capture intricate image features, while concurrently delving into the nuanced chemical structural information across multiple hierarchical levels of molecules. This holistic approach enables our model to generate a significantly more precise and comprehensive representation of molecular features.

The novelty of this research underscores three contributions: firstly, a pioneering approach to molecular data representation is introduced through the novel concept of graph structuring molecular images. This perspective offers a novel and refreshing dimension to data handling, enriching the methodological landscape in the field. Secondly, a novel fusion training paradigm is formulated, seamlessly integrating a graph learner with branches of anchor views and learner views, all within the robust BYOL [8] (Bootstrap Your Own Latent) self-supervised learning framework. Additionally, tailored molecular data augmentation techniques are developed, enabling bootstrapping learning of molecular chemical structures and ultimately resulting in a comprehensive and accurate molecular representation. Thirdly, an innovative graph attention mechanism, rooted in weighted graph isomorphism networks, is proposed for the processing of graph-structured data. This mechanism gracefully adjusts its focus across varying levels of information, spanning from superpixel fragments to individual atoms, chemical bonds, and ultimately functional groups. This approach mirrors the intuitive progression of image understanding, transitioning from fine-grained analysis to holistic comprehension [9]. Importantly, it incorporates inductive biases into the multi-level self-attention framework, effectively mitigating complexity and alleviating training challenges, thereby enhancing the overall model performance and interpretability.

The subsequent sections of this paper are meticulously structured as follows: Initially, the upcoming section presents a comprehensive review of related work. Following this exhaustive overview, Section 3 delves

profoundly into the realm of Molecular Image Representation Learning, encompassing the Self-supervised Learning framework, intricate routing attention mechanisms, graph attention mechanisms intertwined with Graph Isomorphism Networks, and the graph-pooling approach. Subsequently, Section 4 showcases the rigorous experimental setup, detailing the validation process that underscores the effectiveness of our proposed approach. Ultimately, the paper concludes with a concise summary, accompanied by a thoughtful discussion that encapsulates our key findings and their implications, fostering deeper insights into the field.

2. Related work

In the realm of molecular representation learning, methodologies are primarily categorized by the nature of input molecular representations: **sequence-based** and **graph-based** approaches. While both have advanced the field, self-supervised learning (SSL) has emerged as a critical frontier for leveraging unlabeled data, with recent efforts like MolCLR and ImageMol providing foundational insights.

Sequence-based methods: given that molecular structures can be conceptualized as linear sequences (encompassing SMILES strings, InChI expressions, among others), this research domain delves into a spectrum of deep learning frameworks tailored specifically for sequence data. Examples include SmilesLSTM [10], Transformer models [11], and Variational Autoencoders [12], which are leveraged to capture intricate long-range dependencies within these sequences. The primary objective of these methods is to extract meaningful chemical information, thereby facilitating the efficient analysis and prediction of molecular properties. However, these methods inherently struggle to model the topological complexity of molecular structures, as linear sequences obscure spatial relationships between atoms and functional groups.

Graph-based methods: The research within this segment delves into Graph Neural Networks (GNNs) and their variants, particularly Message Passing Neural Networks (MPNNs) [7] and Graph Attention Networks (GATs) [13], to directly learn representations of nodes and edges embedded within molecular graph structures. This approach facilitates a more intuitive and precise simulation of intermolecular interactions and properties, since graph representations endow the models with enhanced capabilities to comprehend the intricate internal structures and attributes of molecules. For instance, the AttentiveFP [14] model harnesses an attention mechanism to aggregate node information, thereby generating a comprehensive graph representation. Moreover, the PremuNet [15] framework innovatively integrates Transformer and GNN modules, empowering it to extract valuable insights from both SMILES molecular expressions and molecular graph data. Nonetheless, traditional GNNs face challenges in hierarchical feature extraction, often treating atoms and bonds as uniform nodes without explicit modeling of functional groups or subgraph motifs.

Self-Supervised Learning (SSL) and Contrastive Approaches: Recent SSL frameworks have sought to address data scarcity by pre-training on unlabeled molecules. For example, MolCLR [6], a leading SSL framework, relies on predefined graph augmentations such as atom masking and subgraph removal. While effective, these static strategies can lead to overfitting—especially under overly aggressive augmentation that removes critical substructures, or in tasks sensitive to specific molecular topologies (e.g., BBBP), where such distortions may impair generalization. ImageMol [3], another SSL method, treats molecules as images and uses CNNs for feature extraction but lacks integration of graph-structured semantics, leading to limited explicit modeling of chemical semantics, such as atomic properties and bond types. Unlike graph-based methods that can directly encode atomic charges, bond orders, and functional group information into node/edge features, ImageMol extracts features from pixel-level molecular images, which may implicitly capture structural patterns but lacks direct integration of chemical domain knowledge.

Our proposed method MolRL distinguishes itself by bridging molecular images and graph structures through structure bootstrapping

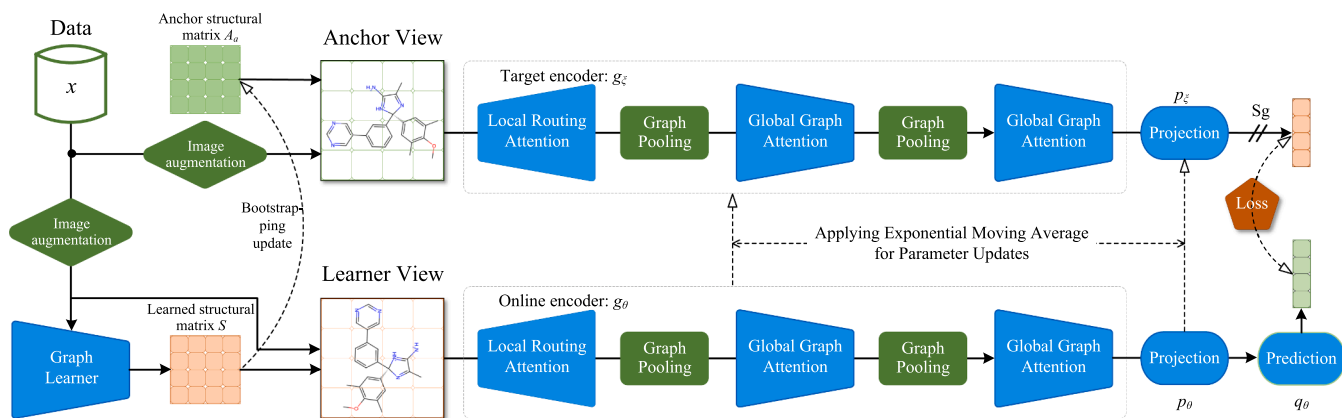


Fig. 1. Pipeline for Molecular Image Representation Learning, wherein S represents the learned structural matrix derived from the segmented molecular image, and A designates the initial spatial adjacency matrix. θ encapsulates the trained weights of the networks, whereas ξ signifies the exponential moving average of these weights. The notation Sg indicates the application of the stop-gradient operation, which halts gradient propagation through the target branch of the network during backpropagation.

self-supervision, a paradigm absent in prior work. Unlike conventional graph-based self-supervised learning (SSL) models—represented by MolCLR [6]—which rely on static graph augmentations and predefined topological priors, MolRL introduces three key innovations: 1) Integrate CNN and GNN strengths: by dissecting molecular images into patches (as graph nodes), MolRL uses CNNs to extract local visual features (e.g., atomic edges, bond textures) and hierarchical GNNs with routing and global attention to model chemical semantics. 2) Hierarchical attention mechanism: MolRL’s weighted GIN with routing attention focuses on local substructures (e.g., halogenated rings, carbonyl groups) while global attention aggregates multi-scale features. This contrasts with MolCLR’s flat graph attention, which struggles to disentangle functional group contributions. 3) Structure-guided pre-training: by iteratively updating anchor graph structures using learned adjacency matrices, MolRL avoids overfitting to fixed topologies, unlike MolCLR’s static graph augmentations. This adaptability enables MolRL to generalize better to diverse molecular families.

3. Molecular image representation learning

In the methodological pipeline, molecular images first undergo an initial dissection into distinct patches. Subsequently, a convolutional token embedding mechanism is employed to model the local contexts within these molecular patches, utilizing a single-layer convolutional network. Specifically, when considering a 2D image as input, we learn a function $f(\cdot)$ that proficiently maps each patch into a unique token. Following this tokenization process, the tokens, subsequently viewed as graph nodes, engage in a paradigm of graph structure bootstrapping self-supervision.

3.1. Structure bootstrapping self-supervised learning

3.1.1. Overall framework logic

The structure bootstrapping self-supervised learning framework aims to learn robust molecular representations by leveraging dual views of molecular images and dynamically updating graph structures. As illustrated in Fig. 1, the framework consists of three core components: 1) Dual-View Construction: generates anchor and learner views via distinct data augmentations and graph structure matrices; 2) Asymmetric Neural Network Architecture: employs an online network for learner view processing and a target network for stable representation generation; 3) Contrastive Loss and Optimization: encourages representation consistency between views while iteratively refining graph structures.

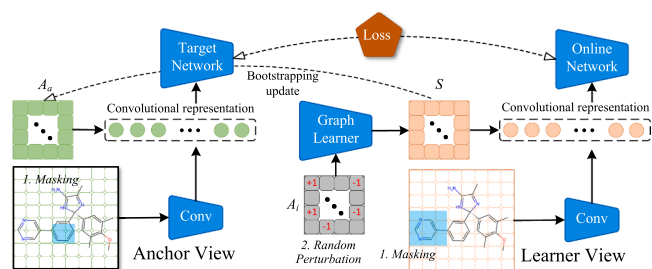


Fig. 2. Illustrative representation of two types of data augmentation techniques.

3.1.2. Dual-view construction

To enable contrastive learning, the framework creates two correlated yet distinct views of the same molecular image by transforming molecular image samples into an anchor view and a learner view through distinct augmentation strategies and input graph structure matrices (as shown in Fig. 1). Specifically, the anchor view structure matrix is initialized as an eight-direction spatial adjacency matrix A_i to capture the inherent spatial relationships. However, maximizing the consistency between two views often leads to overfitting of the learned graph structure to the fixed anchor view structure, hindering the acquisition of an accurate graph representation. To tackle this issue, we introduce a structure guidance mechanism. At the core of this mechanism lies the gradual updating of the anchor view structure A_a , guided by the insights derived from the dynamically learned graph structure S . Given a decay rate τ , we have:

$$A_a \leftarrow \tau A_a + (1 - \tau)S. \quad (1)$$

The complete pseudocode for this dynamic graph structure update mechanism is provided in Appendix Algorithm 1. Specifically, τ regulates the contribution of the historical anchor view structure, ensuring stable iterative updates of A_a ; meanwhile, gradient propagation from the anchor view to the learner view is blocked (via gradient detach operations) to avoid training instability caused by cross-view gradient interference; the update of A_a is performed every 3 iterations to balance the preservation of prior structural information from the anchor view and the integration of new insights from the dynamically learned S .

Regarding data augmentation, we employ two strategic techniques on input molecular images (see Fig. 2):

1) **Molecular Image Masking.** We apply randomly masking technique to the images, training the neural network to accurately predict and reconstruct the missing atom, bond or even subgraph of the masked molecular images. This approach strengthens the model’s

capacity to comprehend and interpret complex molecular structures. 2) **Random Perturbation of Graph Structure.** Additionally, we incorporate random perturbations into the initial spatial adjacency matrix input to the graph learner, specifically by changing non-zero elements to zero or vice versa (typically 5–10 % of edges). This challenges the graph learner within an unsupervised framework to eliminate erroneous spatial edge connections and establish meaningful ones, thereby enhancing the model's robustness in handling the initial spatial adjacency matrix and improving graph structural inference capability. The 5–10 % ratio strikes a balance between preserving core molecular connectivity and introducing topological ambiguity to challenge the graph learner.

3.1.3. Neural network architecture

Subsequently, the neural network architecture encompasses three distinct neural networks: a graph learner, an online network, and a target network. Notably, the graph learner adopts an identical neural architecture as the Global Graph Attention component within the encoder network. The online network comprises three stages: an encoder g_θ , a projector p_θ , and a predictor q_θ . The target network, comprising an encoder g_ξ and a projector p_ξ , provides regression targets for training the online network, where its parameters are the exponential moving average of the online network's parameters. This asymmetric structural design can prevent the model from experiencing model collapse during training, which refers to the phenomenon where the two branches of the model produce fixed and identical outputs.

3.1.4. Loss function and optimization

The Mean Squared Error (MSE) loss minimizes the discrepancy between online and target representations:

$$\mathcal{L}_{\theta,\xi} \triangleq \|q_\theta(p_\theta(g_\theta(t(x)))) - p_\xi(g_\xi(t'(x)))\|_2^2, \quad (2)$$

where the symbol $\|\cdot\|_2$ signifies ℓ_2 -norm, t and t' represent transformations for image augmentation on image x . The online network is updated by minimizing the loss function through gradient descent, while at the back end of the target network, gradient calculation is truncated. The updating methods adopted by the two branches are as follows:

$$\theta \leftarrow Adam(\theta, \nabla_\theta \mathcal{L}_{\theta,\xi}, \eta), \quad (3)$$

$$\xi \leftarrow \tau\xi + (1 - \tau)\theta. \quad (4)$$

where ∇_θ denotes the back-propagated gradients, η represents the learning rate, and the *Adam* is an optimizer, the parameters θ undergo updates at each step of the optimization iteration. Conversely, the updates for the parameters ξ are carried out following the completion of each training epoch.

This setup enables the framework to learn robust representations of molecular images in an unsupervised manner, leveraging the inherent structure and properties of the data through the graph learner and diverse views generated by augmentation.

To clarify the interaction between the two views and how contrastive learning ensures their representation consistency, we further elaborate on the asymmetric network design and loss optimization: 1) the learner view is processed by the online network, which dynamically learns features from augmented molecular images and outputs predicted representations. The anchor view is handled by the target network, generating stable reference representations; critically, target network parameters ξ are updated as the exponential moving average of online network parameters (Eq. (3)), ensuring stable reference signals. 2) contrastive learning enforces representation consistency via Mean Squared Error (MSE) loss, which minimizes the L_2 -norm discrepancy between the learner view's predicted representations and the anchor view's reference representations. 3) During training, the online network is optimized by minimizing this loss (updating θ via gradient descent), while gradient propagation in the target network is truncated to avoid instability. This design, coupled with the structure guidance mechanism (Eq. (1)), forms a closed loop: the learner view's dynamically learned S refines the anchor view's A_a , and the MSE loss aligns their feature distributions, jointly ensuring structural and representational consistency.

3.2. Hierarchical attention via graph isomorphism networks

3.2.1. Weighted graph isomorphism networks

Graph Isomorphism Networks (GINs) [16] are a type of neural network model designed specifically for graph representation learning. Their core motivation stems from the imperative for Graph Neural Networks (GNNs) to embody the injective property, a quality that guarantees unique representations for distinct graph structures. GINs achieve this by incorporating a learnable parameter ϵ that adjusts the importance of a node's own features relative to its neighbors' features during aggregation.

In standard GINs, node features are aggregated from their neighbors using a predefined aggregation function (such as sum, max, or mean). Alternatively, our model introduces a weighted node aggregation approach tailored to the continuous values of the learned graph structure matrix S . These values signify the distance and intensity of connections between nodes, contrasting with the binary (0 or 1) representation in conventional models. However, the matrix S , being dense and indicative of a fully connected graph, frequently entails substantial computational costs. Consequently, prior to the message passing phase in graph isomorphism networks, it is imperative to preprocess the matrix through: **I.** Sparsification: refining the matrix by retaining only the top- $\sqrt{N}$ strongest neighbor edges per node, where N denotes the total number of nodes. This corresponds to setting the sparsity parameter $\lambda = \frac{1}{\sqrt{N}}$, such that each node retains $\frac{N}{\sqrt{N}} = \sqrt{N}$ edges with the highest weights. This strategy reduces the computational complexity from $O(N^2)$ to $O(N\sqrt{N})$, promoting computational efficiency while focusing on the most significant structural connections within the graph. **II.** Symmetrization and normalization:

$$\tilde{S} = \frac{S^{(sp)} + (S^{(sp)})^T}{2} \rightarrow S = \tilde{D}^{-\frac{1}{2}} \tilde{S} \tilde{D}^{-\frac{1}{2}}, \quad (5)$$

where $S^{(sp)}$ represents matrix after sparsification, $\tilde{D}$ is a degree matrix of $\tilde{S}$. The A_a matrix of anchor view applies the same preprocessing procedure because it adopts S for bootstrapping update. This symmetrization and normalization process ensures the graph structure matrix S exhibits symmetric properties, whose eigenvalues are real numbers and facilitate stable gradient propagation in deep networks. By constraining the spectral radius of S via degree matrix normalization (analogous to GCN's symmetric normalization strategy [17]), the model mitigates gradient vanishing/exploding issues caused by node degree disparities, thereby promoting convergence stability during training.

The refined formula for updating node features is expressed as follows:

$$f_v^{(\ell)} = MLP^{(\ell)}\left((1 + \epsilon^{(\ell)}) \cdot f_v^{(\ell-1)} + \sum_{u \in \mathcal{N}(v)} e_{u,v} \cdot f_u^{(\ell-1)}\right), \quad (6)$$

where ℓ denotes the index of layers, *MLP* refers to a multi-layer perceptron consisting of two linear layers and an intermediate activation layer. The term $e_{u,v}$ represents the edge weight between nodes u and v .

3.2.2. Routing attention

Routing attention aims to constrain the scope of attention to graph nodes with higher connectivity scores, as defined by the graph structure matrix, within a given graph data. As depicted in Fig. 1, which features two view branches, in the anchor view, since the graph structure matrix is initialized as a spatial adjacency matrix, it approximately equals the initial value A_i during the initial epochs of training. By selecting the k -nearest neighbors (k -NN) and ensuring $k \geq 8$, it ensures that each node captures its spatially adjacent neighbors in 8 directions, effectively leveraging a 'gather' operation to aggregate the nodes of interest under the attention mechanism. This constitutes a spatial local attention mechanism, facilitating spatial clustering of fragmented molecular image segments and mitigating the challenge of overlooking local crucial information with global attention. Similar processing is applied to the learner view branch, where routing attention focuses on the top k -highest connectivity scores for each node in the S matrix.

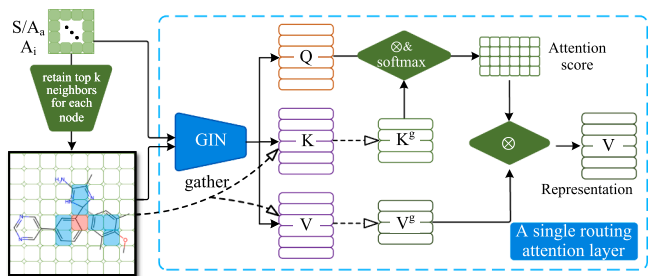


Fig. 3. Illustration of the routing attention process. It employs GIN for KQV computation, and constrains the attention scope by selectively focusing on fragmented molecular image segments with the top k -highest connectivity scores.

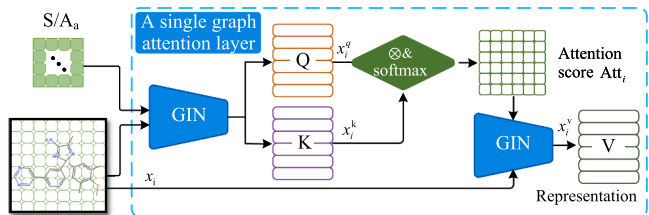


Fig. 4. An illustration of the graph attention process showcases the utilization of GIN for the computation of Key and Query representations, followed by the application of the derived attention scores for the Value computation.

The specific process, as illustrated in Fig. 3, involves utilizing a Graph Isomorphism Network (GIN) to compute the Key, Query, and Value (KQV) representations. Subsequently, node gathering is performed based on the graph structure matrix, and the final graph node feature representation is derived through matrix inner product ($\otimes$). To mitigate the gradient vanishing issue in deep networks, the attention layer incorporates a residual connection structure, where its input is added to its output, ensuring that valuable gradients are effectively propagated throughout the network. This approach harmoniously integrates graph structural information with attention mechanisms, enhancing the model's capability to capture local patterns while preserving contextual integrity.

3.2.3. Graph attention

Graph attention is a global attention mechanism that first leverages the molecular image segmentation map and graph structure matrix, employing the Graph Isomorphism Network (GIN) to compute the Key and Query, thereby generating an attention matrix:

$$Att_i = \text{Attention}(x_i^q, x_i^k) = \text{softmax}\left(\frac{x_i^q (x_i^k)^T}{\sqrt{d}}\right), \quad (7)$$

where d signifies the feature dimension for both the queries and keys. x_i^q and x_i^k represent the sequences of queries and keys, respectively, on the i th attention head. $Att_i \in \mathbb{R}^{n \times n}$ denotes the attention matrix computed specifically for the i th head. Subsequently, GIN is utilized again, based on this generated matrix and the original molecular image segmentation map, to compute the Value (see Fig. 4) by using a residual connection:

$$x_i^v = \text{GIN}(x_i, Att_i) + x_i. \quad (8)$$

where x_i denotes attention module inputs, and x_i^v represents the sequence of values. Notably, the process above incorporates a multi-head mechanism, a choice that can augment the model's representational prowess. However, while increasing the number of heads, it concurrently reduces the head size, thereby diminishing its expressive capability. If the number of heads h exceeds $\frac{d}{n}$, the attention unit within each head will project onto a dimension smaller than n , leading to a low-rank bottleneck that hinders its ability to represent arbitrary context vectors [18].

To address the low-rank bottleneck issue, we adopt the 'talking-heads' approach proposed by Shazeer et al. [19], which incorporates linear projections across the attention-heads dimension. This approach can alleviate the low-rank bottleneck problem, and foster rich interactions among the attention heads, enhancing their collective ability to capture intricate dependencies and nuances within the molecular graph data.

In addition, the Graph Learner in Fig. 1 employs the graph attention structure described here, and the Att_i matrix generated by the final attention layer serves as the learned graph structure input for the subsequent Learner View.

3.2.4. Graph-pooling

As shown in Fig. 1, two graph-pooling layers are utilized between the routing attention layer and the graph attention layer, as well as between the graph attention layers. These pooling layers serve to aggregate molecular image fragments into atomic and chemical bond information, which is further aggregated into functional group information. This hierarchical aggregation of semantic information enables the layered analysis of molecular chemical structures, allowing for a more concise and hierarchical representation of the graph's structural properties.

To achieve this, we propose the use of graph-pooling methods [20]. Given the node feature representations $F = [f_1, f_2, \dots, f_n]^T \in \mathbb{R}^{n \times d}$, which serve as the inputs to each attention module, the formulation of second-order pooling can be expressed as: $F' = U F^T F \in \mathbb{R}^{t \times d}$. Here, $U \in \mathbb{R}^{t \times t}$ is a transformation matrix. Regarding the graph structural matrix S or A_a , which are pertinent to the learner view and anchor view respectively, we derive $C = U F^T \in \mathbb{R}^{t \times n}$, where C serves as a transformation of the input features. Subsequently, the new structural matrix S' or A'_a is computed as $S'/A'_a = C(S/A_a)C^T \in \mathbb{R}^{t \times t}$, which captures the restructured relationships among the aggregated nodes in a reduced dimensionality space.

4. Experiments

4.1. Experimental setup

The pipeline for molecular image representation learning, as illustrated in Fig. 1, integrates a Graph Learner and an encoder. This encoder consists of three key modules: a routing attention module, two graph attention modules, and has a total of 117 million parameters. Each module comprises four attention residual blocks, featuring a hidden size of 512. The Weighted Graph Isomorphism Networks module meticulously controls the sparsity of the graph structure matrix via a λ parameter, tailored as $n\sqrt{n}$, where n represents the count of graph nodes. The routing attention module employs an efficient k -nearest neighbor approach, with k set to $\sqrt{n}$. Molecular images fed into the model undergo normalization to a uniform 240×240 pixel resolution, utilizing a patch size of 15, thereby generating 256 input graph nodes. Subsequently, a hierarchical graph pooling strategy is implemented: the first layer retains 64 graph nodes, while the second layer further refines this to 16 representative nodes. This streamlined neural network architecture was pre-trained on a setup consisting of four NVIDIA A100 Tensor Core GPUs for a total of 600 epochs, utilizing the Adam optimizer with an initial learning rate of $1e^{-4}$ that was subjected to decay over the course of training. For the pre-training dataset, we leveraged approximately 10 million unlabeled molecule SMILES strings from the database gathered by ChemBERTa [21] from PubChem. Subsequently, these SMILES strings were converted into molecular images using RDKit.

4.2. Molecular property predictions

4.2.1. Classification tasks

To demonstrate the effectiveness of MolRL, we benchmark the performance on multiple challenging classification and regression tasks from MoleculeNet [22]. For all datasets, excluding QM9, we adhere to the scaffold splitting strategy advocated in [5,23] to devise an 80/10/10

Table 1
Evaluating and Comparing Test Performance of Diverse Models Across Seven Classification Benchmarks.

Dataset # Molecules # Tasks	BBBP 2039 1	Tox21 7831 12	ClinTox 1478 2	HIV 41127 1	BACE 1513 1	SIDER 1427 27	MUV 93087 17
GCN [17] (2017)	71.8 ± 0.9	70.9 ± 2.6	62.5 ± 2.8	74.0 ± 3.0	71.6 ± 2.0	53.6 ± 3.2	71.6 ± 4.0
GIN [16] (2019)	65.8 ± 4.5	74.0 ± 0.8	58.0 ± 4.4	75.3 ± 1.9	70.1 ± 5.4	57.3 ± 1.6	71.8 ± 2.5
MGCN [24] (2019)	85.0 ± 6.4	70.7 ± 1.6	63.4 ± 4.2	73.8 ± 1.6	73.4 ± 3.0	55.2 ± 1.8	70.2 ± 3.4
D-MPNN [7] (2019)	71.2 ± 3.8	68.9 ± 1.3	90.5 ± 5.3	75.0 ± 2.1	85.3 ± 5.3	63.2 ± 2.3	76.2 ± 2.8
AttentiveFP [14] (2020)	83.2 ± 1.0	57.6 ± 5.0	94.1 ± 5.3	--	78.2 ± 5.3	57.6 ± 5.0	--
N-Gram [25] (2019)	91.2 ± 3.0	76.9 ± 2.7	85.5 ± 3.7	83.0 ± 1.3	87.6 ± 3.5	63.2 ± 0.5	81.6 ± 1.9
Hu et al. [23] (2020)	70.8 ± 1.5	78.7 ± 0.4	78.9 ± 2.4	80.2 ± 0.9	85.9 ± 0.8	65.2 ± 0.9	81.4 ± 2.0
MAT [26] (2020)	92.2 ± 1.2	80.0 ± 1.8	94.7 ± 6.0	--	84.5 ± 2.1	62.1 ± 2.0	--
MGSSL [27] (2021)	69.7 ± 0.9	76.5 ± 0.3	80.7 ± 2.1	--	79.1 ± 0.9	61.8 ± 0.8	--
GraphMVP [28] (2022)	86.3 ± 5.3	77.5 ± 1.3	56.0 ± 5.8	--	81.3 ± 2.2	60.4 ± 4.8	--
GEM [29] (2022)	89.7 ± 3.7	82.0 ± 0.8	86.2 ± 4.1	--	85.9 ± 5.2	62.6 ± 3.4	--
MolCLR _{GIN} [6] (2022)	73.8 ± 0.2	74.7 ± 0.8	86.7 ± 1.0	77.8 ± 0.5	78.8 ± 0.5	66.9 ± 1.2	84.0 ± 1.8
MolCLR _{GIN} [6] (2022)	73.6 ± 0.5	79.8 ± 0.7	93.2 ± 1.7	80.6 ± 1.1	89.0 ± 0.3	68.0 ± 1.1	88.6 ± 2.2
ImageMol [3] (2022)	73.9 ± 0.2	77.3 ± 0.1	85.1 ± 1.4	79.7 ± 0.2	83.9 ± 0.5	66.0 ± 0.1	82.5 ± 0.8
MOLFORMER-Base [30] (2022)	90.9	43.2	61.3	77.7	82.8	64.8	--
SElFormer [31] (2023)	90.2	65.3	--	68.1	83.2	74.5	--
HiMol [32] (2023)	73.2 ± 0.8	76.2 ± 0.3	80.8 ± 1.4	--	84.3 ± 0.3	61.3 ± 0.5	--
TransFoxMol [11] (2023)	92.5 ± 1.8	80.4 ± 1.0	97.2 ± 2.4	--	86.5 ± 0.5	62.9 ± 2.4	--
MolRL	93.2 ± 2.4	82.5 ± 1.2	94.5 ± 2.1	83.3 ± 0.7	87.7 ± 1.1	68.1 ± 1.3	85.4 ± 1.7
# Rank	(1)	(1)	(3)	(1)	(2)	(2)	(2)

split for training, validation, and testing, respectively. This approach, grounded in molecular substructures, contrasts sharply with the conventional random splitting method, presenting a more arduous yet pragmatic prediction task. Conversely, QM9 conforms to the random splitting convention employed by numerous related works [24,25], enabling a straightforward comparison across studies. The mean and standard deviation of three individual runs are reported.

Table 1 presents the mean and standard deviation of ROC-AUC scores (in percentages) attained on the binary classification tasks, where the first five models are supervised learning methods and the last twelve are self-supervised/pre-training methods. Our key observations from these results are as follows: By incorporating self-supervised pre-training, MolRL demonstrates remarkable robustness, surpassing all the supervised methods and claiming the top position on four of the seven evaluated datasets and securing the second rank on two additional datasets. Even on the remaining dataset, ClinTox, MolRL retains a competitive advantage. MolCLR [6] introduces Molecular Contrastive Learning of Representations via Graph Neural Networks (GNNs), alongside a self-supervised learning framework that capitalizes on vast unlabeled data to pre-train the network. While MolCLR leverages the labeled graph structures inherent in molecular data, it lacks the ability to hierarchically analyze molecular structures. Notably, MolCLR_{GIN} demonstrates outstanding performance on datasets such as BACE and MUV, yet its efficacy falters on other datasets, particularly the BBBP, indicating room for improvement in addressing the complexities of diverse molecular structures. MOLFORMER-Base [30] is a 6-layer Transformer-based molecular language model, utilizing linear attention and rotary positional embeddings, pre-trained on 1.1 billion molecular SMILES sequences via masked language modeling to learn implicit structural patterns from linear text. In performance, MOLFORMER-Base achieves an AUC of 90.9 on BBBP and 61.3 on ClinTox versus MolRL's 93.2±2.4 and 94.5±2.1 respectively, with MolRL's image-to-graph framework demonstrating advantages in capturing spatial and topological information for complex tasks. SElFormer [31] leverages a Transformer architecture with byte-level BPE tokenization, pre-trained via masked language modeling on 2 million drug-like molecules in SELFIES format, and demonstrates competitive performance, achieving an AUC of 83.2 on BACE and 74.5 on SIDER, outperforming MOLFORMER-Base (82.8 on BACE, 64.8 on SIDER) and other graph-based models on specific tasks like SIDER, while showing comparable results on others. TransFoxMol [11] introduces an elegant transformer-based framework with focused attention for molec-

ular representation, showcasing that utilizing attention mechanisms to enhance machine understanding of molecular structures is a significant approach in molecular representation learning. However, this method relies on prior knowledge such as chemical maps and lacks the generative learning capability for molecular graph structures. While TransFoxMol achieves remarkable results on the ClinTox dataset, its performance remains moderate on others.

4.2.2. Regression tasks

The regression outcomes are presented in Table 2. In adherence to the guidelines provided by MoleculeNet, the root-mean-square error (RMSE) has been adopted as the evaluation criterion for physical chemistry datasets, specifically ESOL, FreeSolv, and Lipophilicity. Conversely, for quantum mechanics datasets, which encompass QM7, QM8, and QM9, the mean-absolute error (MAE) serves as the metric of choice. MolRL surpasses all baselines on three out of six datasets, demonstrating remarkable performance, while achieving comparable results on the remaining datasets, highlighting its overall competitiveness. From a comprehensive perspective, the pre-trained MolRL model achieves the best performance, and notably outperforms the strongest baseline by an 8.9% reduction in MAE on the challenging QM9 dataset.

Within this comparative landscape, SElFormer (2023) is cited for its standout performance on the ESOL dataset, achieving an RMSE of 0.682-the best result among compared models. This places our model (MolRL) in third position on this benchmark, highlighting SElFormer's effectiveness in leveraging SELFIES representations and transformer architecture for solubility prediction. SElFormer's superior performance on ESOL can be attributed to its robust handling of molecular string semantics and hierarchical attention mechanisms, whereas MolRL's graph-based framework, while excelling in structural topology modeling, may face challenges in capturing fine-grained chemical feature correlations required for precise aqueous solubility predictions. This comparison underscores the complementary strengths of text-based and graph-based approaches in molecular property prediction.

4.3. Investigation of molecular representation

The feature map heatmaps (Fig. 5) serve as a visualization tool for the intermediate outcomes generated by a model during the feature extraction process. In these heatmaps, warmer hues commonly signify the feature regions that receive heightened attention from the model. The figure demonstrates that MolRL has the capability to adeptly extract

Table 2
Evaluating and Comparing Test Performance of Diverse Models Across Six Regression Benchmarks.

Dataset # Molecules # Tasks	FreeSolv 642 1	ESOL 1128 1	Lipophilicity 4200 1	QM7 6830 1	QM8 21786 12	QM9 130829 8
GCN [17] (2017)	2.87 ± 0.14	1.43 ± 0.05	0.85 ± 0.08	122.9 ± 2.2	0.0366 ± 0.0011	5.796 ± 1.969
GIN [16] (2019)	2.76 ± 0.18	1.45 ± 0.02	0.85 ± 0.07	124.8 ± 0.7	0.0371 ± 0.0009	4.741 ± 0.912
MGCN [24] (2019)	3.35 ± 0.01	1.27 ± 0.15	1.11 ± 0.04	77.6 ± 4.7	0.0223 ± 0.0021	0.050 ± 0.002
D-MPNN [7] (2019)	2.18 ± 0.91	0.98 ± 0.26	0.65 ± 0.05	105.8 ± 13.2	0.0143 ± 0.0022	3.241 ± 0.119
AttentiveFP [14] (2020)	1.838 ± 0.436	0.887 ± 0.068	0.654 ± 0.016	88.73 ± 55.6	--	--
N-Gram [25] (2019)	2.51 ± 0.19	1.10 ± 0.03	0.88 ± 0.12	125.6 ± 1.5	0.0320 ± 0.0032	7.636 ± 0.027
Hu et al. [23] (2020)	2.83 ± 0.12	1.22 ± 0.02	0.74 ± 0.00	110.2 ± 6.4	0.0191 ± 0.0003	4.349 ± 0.061
MGSSL [27] (2021)	2.980	1.346	0.751	155.913	0.0198	21.538
MolCLR _{GCN} [6] (2022)	2.39 ± 0.14	1.16 ± 0.00	0.78 ± 0.01	83.1 ± 4.0	0.0181 ± 0.0002	3.552 ± 0.041
MolCLR _{GIN} [6] (2022)	2.20 ± 0.20	1.11 ± 0.01	0.65 ± 0.08	87.2 ± 2.0	0.0174 ± 0.0013	2.357 ± 0.118
ImageMol [3] (2022)	2.02 ± 0.07	0.97 ± 0.07	0.72 ± 0.01	--	--	3.724 ± 0.025
SElFormer [31] (2023)	2.797	0.682	0.735	--	--	--
HiMol [32] (2023)	2.283	<u>0.833</u>	0.708	91.501	0.0199	3.243
TransFoxMol [11] (2023)	1.77 ± 0.37	0.87 ± 0.04	0.63 ± 0.02	48.6 ± 13.1	--	--
MolRL	1.94 ± 0.21	0.85 ± 0.09	0.61 ± 0.06	75.5 ± 8.2	0.0161 ± 0.0017	2.165 ± 0.483
# Rank	(2)	(3)	(1)	(2)	(1)	(1)

Table 3
Comparative Ablation Analysis Results Across Seven Classification Benchmarks.

Model	Flops	BBBP	Tox21	ClinTox	HIV	BACE	SIDER	MUV
NP-MolRL ^a	67.212G	86.0±5.5	71.3±2.5	88.7±4.9	74.4±1.3	75.6±4.1	61.6±4.2	75.5±3.3
NR-MolRL ^b	80.249G	92.7±3.1	81.4±1.6	93.3±1.8	80.5±0.6	86.1±1.4	66.4±1.7	83.9±2.5
MolRL	67.212G	93.2±2.4	82.5±1.2	94.5±2.1	83.3±0.7	87.7±1.1	68.1±1.3	85.4±1.7

^a NP-MolRL stands for Non-Pretrained MolRL.

^b NR-MolRL stands for NoRouting MolRL.

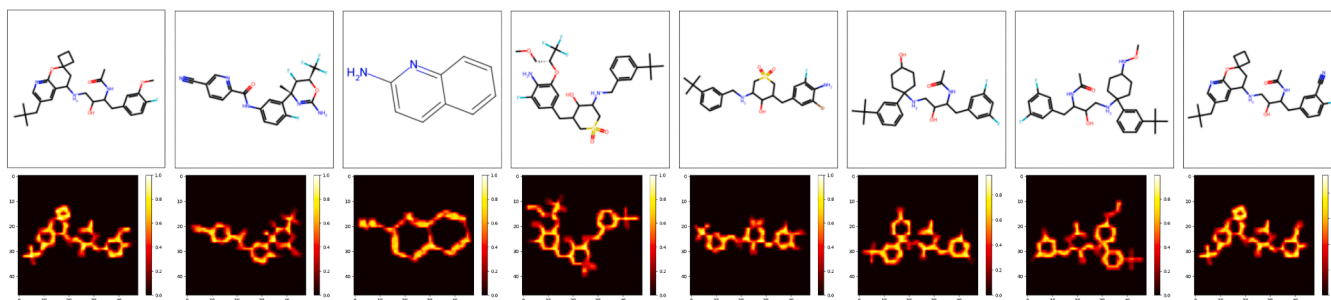


Fig. 5. Low-level feature attention maps from MolRL's initial convolutional layers, highlighting regions corresponding to atoms and chemical bonds in molecular images. These foundational features support hierarchical aggregation of structural information in deeper layers (see Sections 3.2.3-3.2.4 and Figs. 6-7 for higher-order semantic analysis).

features from distinct regions within molecular images, encompassing atoms and chemical bonds, and proficiently captures intricate details during the initial stages of the feature extraction process, which are closer to the input layer. Gradient-weighted Class Activation Mapping (Grad-CAM) is a commonly utilized CNN visualization technique. Fig. 6 presents eight exemplary molecules, showcased through the Grad-CAM visualization technique applied to the self-supervised pre-trained MolRL model. As Grad-CAM is employed on convolutional layers proximal to the input, the generated maps illuminate relatively low-level or localized features within the images. While these features may lack the abstraction and sophistication of those captured by deeper layers, they are nonetheless fundamental for the model to grasp the basic structural and contentual aspects of the input images. Upon examining Fig. 6, it becomes evident that the model has adeptly discerned essential image characteristics, including edges, textures, and colors, all of which are inherent in the molecular graphs and are constructed from atoms and chemical bonds. Notably, the model demonstrates a heightened sensitivity towards chemical bonds that are associated with red oxygen atoms. These distinctive features serve as the foundation for subsequent, more intricate levels of representation learning and property prediction.

Due to the asymmetric dual-view design of our model, where the target network and online network process graph structures dynamically updated from the spatial adjacency matrix and generated by the graph learner, respectively, and since the routing attention mechanism is designed to focus on local graph node connectivity rather than producing a global attention distribution, it is challenging to directly visualize a corresponding attention heatmap over the original molecular image.

To investigate the internal attention mechanism, we instead visualize the attention weight matrix from the final attention layer of the graph learner (see Fig. 7). Specifically, we first compute the row-wise average of the output attention matrix from this layer to aggregate a global measure of how much each query node (image patch) is attended to by all key nodes (image patches). This resulting 1D vector is then reshaped into a 2D square grid to match the spatial layout of the input image patches. Finally, this grid is upsampled to the size of the original molecular image to generate the final heatmap. This heatmap reflects the degree of associative learning between different image patches (nodes) within the model during the graph learner's iterative process of updating the graph structure. Warmer color regions indicate nodes that have received higher global attention weights, signifying that the model

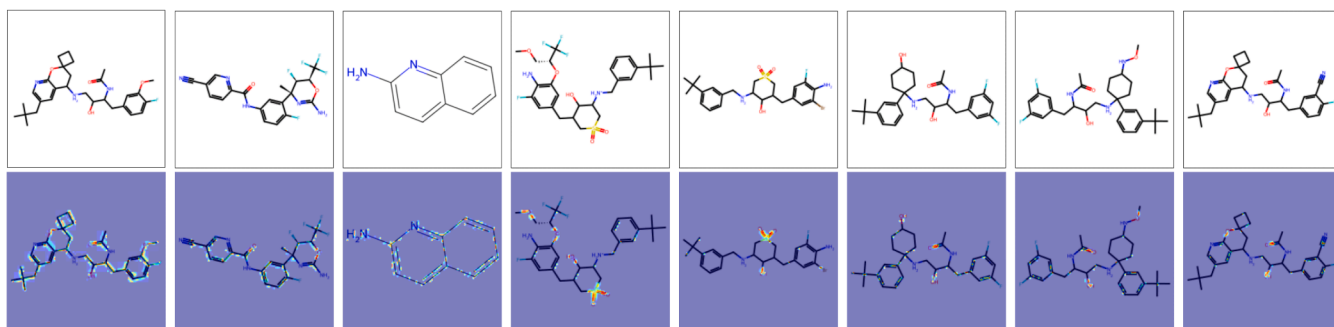


Fig. 6. MolRL’s heatmaps illustrate the focal points of several molecular images, where Grad-CAM highlights the edges and textures. The intensity of focus within each region is reflected by the color gradient, with warmer hues signifying heightened focus and cooler tones indicating lower levels of concentration.

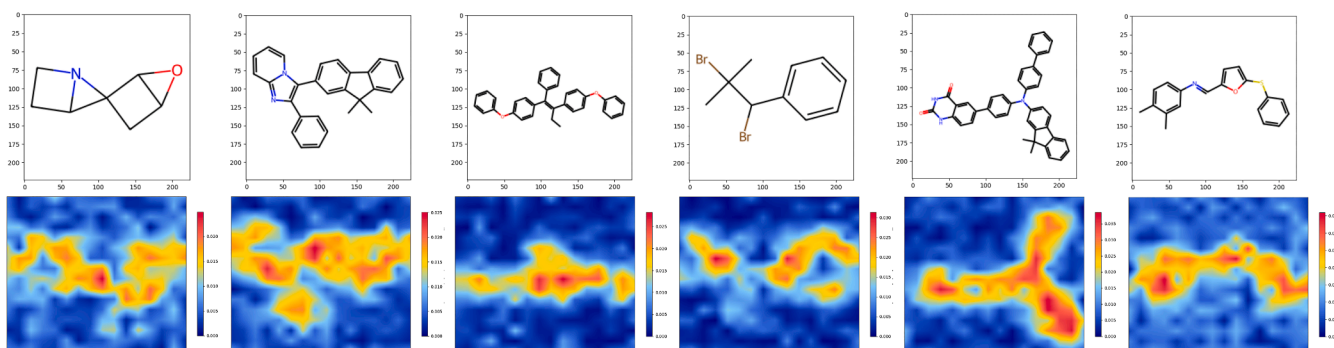


Fig. 7. Heatmap Visualization of Attention Weights from the Last Layer of the Graph Learner.

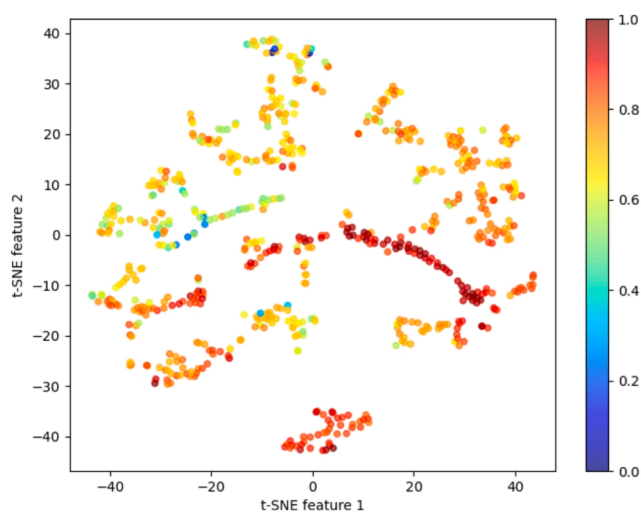


Fig. 8. Molecules from FreeSolv database and color indicates hydration free energy of each molecule.

deems the features of these image patches more critical for the reasoning and evolution of the graph structure. This visualization provides an internal perspective on how the model autonomously learns and evolves graph connectivity from the image data.

Subsequently, we delve into the molecular representations learned by our MolRL pre-training, leveraging the t-SNE embedding technique. The t-SNE algorithm adeptly projects similar molecular representations onto adjacent points within a two-dimensional (2D) space. As depicted in Fig. 8, we visually represent molecule features using t-SNE, where molecules are color-coded according to their respective property labels. Notably, all these features are extracted directly from the pre-trained

Table 4

Classification Accuracy of Functional Group Detection Across Models.

Model	NoRouting MolRL	MolRL	ImageMol
Acc ^a (%)	94.7±0.84	97.6±0.65	89.5±0.62

^a Note: The values represent classification accuracy (%) with mean ± standard deviation across three independent runs.

MolRL model, without any fine-tuning or access to the molecular property labels during the training phase. Features from pre-trained MolRL show clustering based on the labels, even without accessing labels during training. This observation underscores the model’s proficiency in capturing latent representations that reflect the underlying molecular properties.

4.4. Ablation analysis

In this section, we undertake a comparative analysis of two MolRL-based models: Non-Pretrained MolRL and NoRouting MolRL. Specifically, Non-Pretrained MolRL indicates that the model has not undergone large-scale unsupervised pretraining, whereas NoRouting MolRL signifies that the model forgoes the use of routing attention in its bottom layers, opting instead for global attention as a substitute. This nuanced comparison aims to shed light on the distinct representational capabilities of each approach.

First, we delve deeply into the representations acquired by pre-trained versus non-pretrained MolRL models. Fig. 9 showcases the molecules from the validation set of the BACE and BBBP databases, where they have been embedded into 2D using t-SNE and are color-coded according to their respective labels. For comparative purposes, we also include the outcomes from a Non-Pretrained MolRL model. The figure reveals that MolRL, particularly when preceded by large-scale

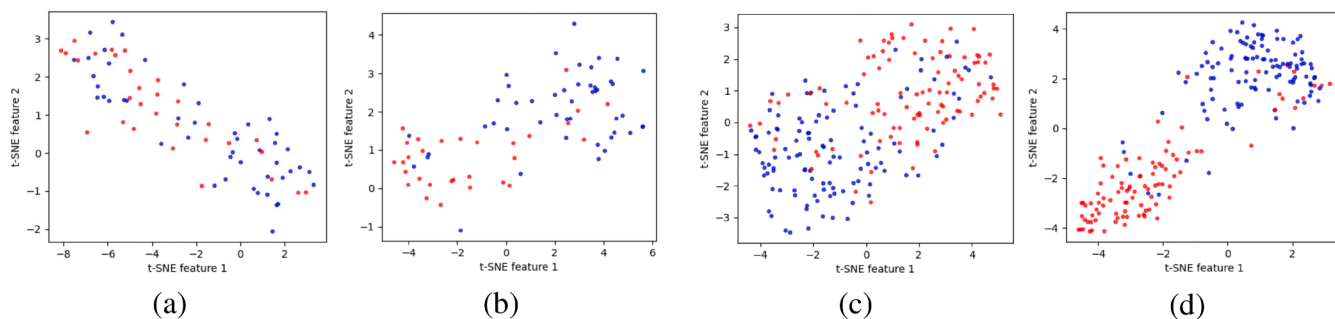


Fig. 9. t-SNE visualization of feature spaces for two validation sets: (a) Non-Pretrained MolRL applied to BACE dataset, (b) MolRL on BACE, (c) Non-Pretrained MolRL applied to BBBP dataset, (d) MolRL on BBBP, with color indicating molecular permeability within each dataset.

Table 5

Comparison of different methods on KIBA and Davis datasets.

Dataset	Methods	MSE	Methods	MSE
KIBA	GraphMVP [28]	0.178±0.005	MGraphDTA [35]	0.128±0.001
	Co-VAE [36]	0.265±0.0014	TransVAE-DTA [37]	0.254±0.0076
	GraphDTA [34]	0.482±0.0081	AttentionMGT-DTA [38]	0.140
	HiSIF-DTA ^a [39]	0.120	MolRL-DTA (Ours)	0.116±0.002
Davis	GraphMVP [28]	0.280±0.005	MGraphDTA [35]	0.207±0.001
	Co-VAE [36]	0.335±0.0136	TransVAE-DTA [37]	0.333±0.0075
	GraphDTA [34]	0.301±0.0054	AttentionMGT-DTA [38]	0.193±0.010
	HiSIF-DTA ^a [39]	0.191	MolRL-DTA (Ours)	0.168±0.004

^a The standard deviation for HiSIF-DTA was not reported in its original paper [39].

self-supervised pre-training, is capable of learning representations that are notably closer for molecules sharing the same label, surpassing the cohesion achieved by its non-pretrained counterpart. The comparative results are presented in Table 3, where MolRL improves largely over NP-MolRL due to the large-scale self-supervised pre-training.

Next, NoRouting MolRL (NR-MolRL) eliminates the routing attention mechanism within the neural network’s foundational layer, prompting the network to prematurely attend to global information during the integration of atomic and chemical bond fragments within molecules. This early shift in focus results in the loss of the inductive bias towards local texture, subsequently impacting the efficacy of the attention mechanism. As evident from Table 3, the performance of NR-MolRL marginally trails that of MolRL, while the standard deviation across three independent runs is slightly elevated, suggesting a slight decrement in the stability of NR-MolRL’s performance. Notably, the computational cost of NR-MolRL is significantly higher than MolRL’s: NR-MolRL requires 80.249G FLOPs, 19.4% more than MolRL’s 67.212G FLOPs. This discrepancy arises because global attention operates on fully connected graphs ($O(N^2)$ complexity), while MolRL’s routing attention reduces connectivity to $O(N\sqrt{N})$ via Structured Sparsification. Despite the higher computational load, NR-MolRL’s performance remains inferior, underscoring that routing attention is essential for efficiently leveraging computational resources to capture both local and global molecular features. In contrast, MolRL achieves superior performance with lower FLOPs, validating the efficiency of its hierarchical attention design in balancing structural detail and computational economy.

To further validate the role of routing attention in capturing fine-grained substructural information, we extend this ablation study by introducing a functional group detection task—focusing on six key functional groups (Hydroxyl: $-OH$, Carbonyl: $C=O$, Amino: $-NH_2/-NHR/-NR_2$, Aromatic Ring: e.g., benzene C_6H_6 , Cyano: $-CN$, Thiol: $-SH$)—to compare MolRL and NoRouting MolRL directly (see Fig. 10). This task specifically aims to quantify how routing attention (a core component of MolRL’s bottom layers) enhances the model’s ability to identify functional groups, as routing attention constrains

focus to locally connected graph nodes (molecular image patches) and mitigates the dilution of local structural signals seen in NoRouting MolRL’s global attention. We additionally include ImageMol [3] as a baseline for this task, as it is the only existing self-supervised framework that leverages molecular images (rather than SMILES-derived graph structures) for representation learning. This comparison is critical because it isolates the model’s capability to recognize functional groups from image-based features (e.g., edge textures, color-coded atoms in molecular images) rather than relying on explicit structural cues from SMILES—capabilities that cannot be replicated by rule-based tools like RDKit (which identifies functional groups via predefined SMARTS patterns, not visual semantics).

To ensure the task focuses strictly on functional group-specific recognition and avoids interference from mixed substructural signals, we construct a specialized dataset where each molecule contains exactly one of the six target functional groups (i.e., the groups are mutually exclusive). We curate 10,000 molecular images per functional group category: 8,000 images for the training set, 1000 images for the validation set, and 1000 images for the test set.¹ All samples are derived from filtered subsets of the PubChem database.

As presented in Table 4, MolRL achieves the highest classification accuracy of $97.6\pm0.65\%$, outperforming NoRouting MolRL ($94.7\pm0.84\%$) by 2.9 percentage points. This advantage validates the critical role of routing attention in capturing fine-grained functional group features: by constraining attention to locally connected molecular image patches, it avoids the dilution of local structural signals that occurs under NoRouting MolRL’s global attention. Notably, ImageMol lags behind both MolRL variants with an accuracy of $89.5\pm0.62\%$. This further underscores the advantage of MolRL’s hierarchical image-to-graph design over alternative methods in extracting chemically meaningful substructures from visual molecular data.

4.5. Drug-target affinity (DTA) prediction

Drug-target affinity (DTA) prediction is a critical task in drug discovery, aiming to model molecular drugs and target proteins to forecast their binding affinity scores [33]. Contemporary approaches like GraphDTA [34] and GraphMVP [28] typically employ graph neural networks (GNNs) for molecular representation learning while utilizing convolutional neural networks (CNNs) to process protein sequences. Building upon this paradigm, we implement our MolRL framework as a graph-based pre-training module to enhance molecular feature extraction for DTA prediction. The KIBA and Davis datasets serve as established benchmarks for evaluating DTA models, with Table 5 comparing our approach against state-of-the-art methods using metrics from recent comprehensive evaluations [33]. Our experiments follow GraphMVP’s

¹ The dataset used in this study is publicly available for download at: https://pan.baidu.com/s/1TX_GoFeQOOW4bDfhYIS23g?pwd=8f9t

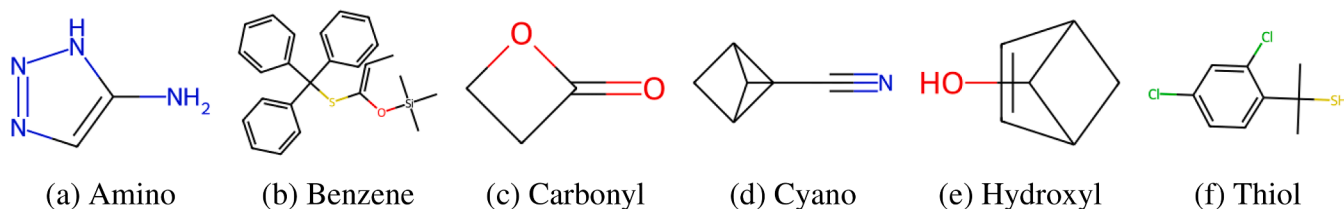


Fig. 10. Representative Molecules for Each of the Six Functional Groups: Amino ($-NH_2/-NHR/-NR_2$), Aromatic Ring (benzene structure as an example), Carbonyl ($C=O$), Cyano ($-CN$), Hydroxyl ($-OH$), and Thiol ($-SH$).

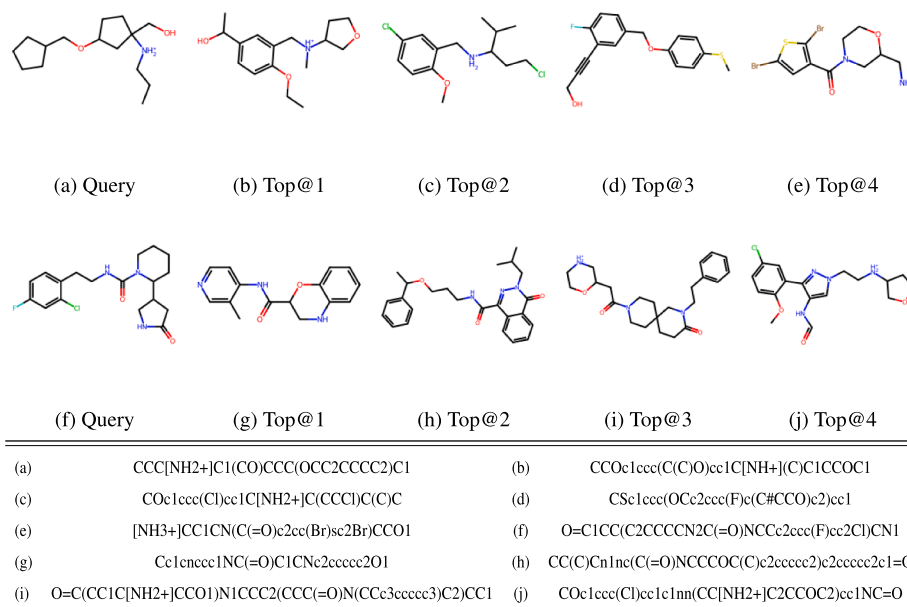


Fig. 11. Visualization of the top-four molecules ranked by molecular representation similarity for the two queries. SMILES for all molecules are given.

protocol [28], implementing random data splitting with three seeds and reporting the mean MSE across three independent runs to ensure statistical reliability.

As shown in Table 5, MolRL-DTA achieves state-of-the-art performance on both datasets, attaining MSE values of 0.116 (KIBA) and 0.168 (Davis). This superiority stems from MolRL’s structure-guided pre-training, which captures nuanced atomic interactions more effectively than sequence-focused approaches like TransVAE-DTA (0.254/0.323). Notably, MolRL consistently outperforms vanilla GNN baselines across multiple runs (e.g., GraphMVP: 0.178 ± 0.005 on KIBA, 0.280 ± 0.005 on Davis), demonstrating the efficacy of our hierarchical attention mechanism in modeling multi-scale chemical features with improved stability and generalization. The consistent improvements across datasets validate MolRL’s ability to bridge 2D structural semantics with 3D binding dynamics, providing a robust framework for molecular representation learning in drug discovery.

4.6. Molecular representation similarity ranking

To delve deeper into the semantic information embedded in the pre-trained molecular representations via MolRL, we assess the similarity between the query molecule and other molecules and present the top-four most similar molecules. The cosine similarity between the query molecule q and the candidate molecule c is computed as follows:

$$\text{sim}(q, c) = \frac{\mathbf{F}^q \cdot \mathbf{F}^c}{\|\mathbf{F}^q\| \|\mathbf{F}^c\|}, \quad (9)$$

where $\mathbf{F}^q \in \mathbb{R}^d$ and $\mathbf{F}^c \in \mathbb{R}^d$ denote the feature representations of the query molecule and candidate molecule, respectively. Subsequently, the similarities of all candidate molecules are ranked, and the top-four molecules for each query are presented in Fig. 11.

From the figure, we can observe that the top-four molecules for each query share notable structural and chemical similarities with the respective query molecules. Specifically, all these molecules contain common atom types such as C, H, N, and also including O. Notably, all similar molecules exhibit shared structural patterns like fused or connected rings, which are critical for molecular properties. The Top@1 molecules in both cases not only mimic the query’s ring architectures but also preserve key chain structures (e.g., $NC(=O)$, $C(=O)N$) that are indicative of functional group interactions. The molecular representation similarity ranking experiments imply that our framework MolRL can learn rich molecular representations embedding chemical semantic information, such as functional groups and aromatic rings. By capturing these structural and functional similarities, the model effectively encodes chemical substructures, which are essential for tasks like molecular property prediction and drug discovery. This capability underscores the framework’s proficiency in extracting meaningful chemical information from molecular images, facilitating accurate similarity assessments based on underlying structural and semantic features.

5. Conclusions

In conclusion, this study presents MolRL, a novel self-supervised learning framework that effectively bridges molecular image represen-

tation with graph-structured reasoning through graph-based bootstrapping and hierarchical attentive graph isomorphism networks. The model demonstrates a strong capacity to learn informative and transferable molecular representations directly from unlabeled image data, enabling robust performance across a diverse range of molecular property prediction tasks. Key innovations include the integration of convolutional feature extraction with graph-based semantic learning, the use of routing attention to capture local substructures, and graph pooling for hierarchical aggregation from image patches to functional groups. These capabilities allow MolRL to achieve state-of-the-art or highly competitive results on multiple classification and regression benchmarks, confirming its generalizability and representation power. Despite its strengths, the framework faces limitations in computational efficiency, interpretability of the hybrid architecture, and dependence on 2D molecular depictions that may not fully capture stereochemical complexity. Nonetheless, the work offers significant value to the research community by providing an open-source, pre-trainable model that reduces reliance on labeled data and serves as a foundation for drug discovery, chemical property prediction, and materials science. Future work will focus on integrating 3D structural information, developing sparse attention to reduce computational complexity, incorporating external chemical knowledge, and extending to generative tasks like de novo molecule design. Overall, MolRL effectively combines visual and structural learning principles to provide a powerful, flexible framework for molecular representation learning, marking a meaningful step toward more general and expressive AI systems for molecular science.

CRedit authorship contribution statement

Dongjing Shan: Writing – original draft, Software, Methodology, Conceptualization; **Yamei Luo:** Writing – original draft, Methodology, Funding acquisition; **Hong Yuan:** Visualization, Software, Data curation; **Jiashun Mao:** Software, Resources, Project administration, Data curation; **Limin Wang:** Writing – original draft, Supervision, Methodology.

Data availability

Data will be made available on request.

Declaration of interests

None declared.

Appendix A. Supplementary Pseudocode for Core Mechanisms

This appendix presents supplementary pseudocode for the dynamic graph structure update mechanism in the MolRL framework (Algorithm 1).

Algorithm 1 Contrastive Learning with Structure Guidance.

Require: Dataset $\mathcal{D}$, augmentation strategies $\mathcal{T}_a, \mathcal{T}_l$, decay rate τ , update interval T

Ensure: Learned representations and graph structures

```

1: Initialize online network  $f_o$  (with encoder  $g_\theta$ , projector  $p_\theta$ , predictor  $q_\theta$ ) and target network  $f_t$  (with encoder  $g_\xi$ , projector  $p_\xi$ ) with weights  $\theta, \xi$ 
2: Initialize anchor structure matrix  $A_a$  as 8-direction spatial adjacency

3: for iteration = 1 to max_iterations do
4:   for each batch  $(I_1, I_2) \sim \mathcal{D}$  do
5:     // Dual-View Construction
6:     Anchor view:  $\tilde{I}_a \leftarrow \mathcal{T}_a(I_1)$  with structure  $A_a$ 
7:     Learner view:  $\tilde{I}_l \leftarrow \mathcal{T}_l(I_2)$  with structure  $S$ 
8:     // Asymmetric Processing
9:     Learner representation:  $z_l \leftarrow q_\theta(p_\theta(g_\theta(\tilde{I}_l, S)))$ 
10:    Target representation:  $z_a \leftarrow p_\xi(g_\xi(\tilde{I}_a, A_a))$   $\triangleright$  Stop gradient
11:    // MSE Loss Calculation
12:     $\mathcal{L} \leftarrow \|z_l - z_a\|_2^2$ 
13:    Update  $\theta_o$  via  $\nabla_{\theta_o} \mathcal{L}$ 
14:    // Structure Guidance Update
15:    if iteration mod  $T = 0$  then
16:       $A_a \leftarrow \tau \cdot A_a + (1 - \tau) \cdot S$   $\triangleright$  EMA update
17:    end if
18:  end for
19:  // Target Network Update
20:   $\theta \leftarrow \mu\theta + (1 - \mu)\xi$   $\triangleright$  Momentum update
21: end for

```

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